

Linear algebra – 10

Matrix norms and localisation of the spectrum

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Theoretical minimum.

Def. A norm on M_n is *submultiplicative* if $\|AB\| \leq \|A\| \|B\|$. The *operator norm* subordinate to a vector norm is $\|A\| = \max_{\|x\|=1} \|Ax\|$; the *spectral radius* is $\rho(A) = \max\{|\lambda| : \lambda \in \text{Spec } A\}$.

Theorem 1 (Gelfand formula). $\rho(A) = \lim_{k \rightarrow \infty} \|A^k\|^{1/k}$ for every submultiplicative norm; consequently $A^k \rightarrow 0$ if and only if $\rho(A) < 1$.

Theorem 2 (Gershgorin 1931). Every eigenvalue of $A = (a_{ij})$ lies in one of the discs $\{z : |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}|\}$; a connected component consisting of k discs contains exactly k eigenvalues counted with multiplicity.

Theorem 3 (Courant–Fischer). For a Hermitian A with $\lambda_1 \geq \dots \geq \lambda_n$,

$$\lambda_k = \max_{\dim L=k} \min_{x \in L \setminus \{0\}} \frac{\langle Ax, x \rangle}{\langle x, x \rangle}.$$

Hence the eigenvalues of a principal submatrix interlace those of A (Cauchy).

1. Prove Gelfand's formula, and deduce that $\rho(A) \leq \|A\|$ for every submultiplicative norm and that for every $\varepsilon > 0$ there is a norm with $\|A\| \leq \rho(A) + \varepsilon$.
2. (VJIMC 2003, cat. II) Two real square matrices A and B satisfy $A^{2002} = B^{2003} = E$ and $AB = BA$. Prove that $A + B + E$ is invertible. (Hint: commuting matrices are simultaneously triangularisable, and the eigenvalues lie on the unit circle.)
3. (IMC) For a real $n \times n$ matrix let $\|M\| = \sup_{x \neq 0} \|Mx\|_2 / \|x\|_2$. Assume that $\|A^k - A^{k-1}\| \leq \frac{1}{2002k}$ for all positive integers k . Prove that $\|A^k\| \leq 2002$ for all k .

4. (IMC) Let A be a complex $n \times n$ matrix all of whose eigenvalues have modulus at most 1. Prove that $\|A^n\| \leq \frac{n}{\ln 2} \|A\|^{n-1}$.
5. (SEEMOUS 2011) Let $A = (a_{ij})$ be a real $n \times n$ matrix with $A^n \neq O_n$ and $a_{ij}a_{ji} \leq 0$ for all i, j . Prove that A has at least two non-real eigenvalues.